Mahesh Pulivarthi

Portfolio: my-me-one.vercel.app | GitHub: MaheshPulivarthi18 | LinkedIn: in/mahesh-pulivarthi | Email: maheshpulivarthi18@gmail.com

Profile

Backend / Platform Engineer with 1.5+ years of experience building production wealthtech systems across microservices, distributed systems, and multi-database architectures. Experienced with Python, JavaScript, Go, Kubernetes, Kafka, Redis, and OpenTelemetry, with a focus on scalable backend systems, reliability, and observability.

Experience

Valuefy Jan 2025 – Present

Software Engineer (Jun 2025 – Present) | Software Engineer Intern (Jan 2025 – Jun 2025)

- Owned an investment management module end-to-end, designing backend services, data flows, and production deployments, including master-data synchronisation using incremental batch processing.
- Built an AI-powered statement extraction pipeline using FastAPI, OCR, LLM structured outputs, and Pydantic validation, turning scanned statements into structured investment data.
- Designed a database-agnostic persistence layer so the same wealth-management platform runs on PostgreSQL, MySQL, and Oracle without client-specific code forks.
- Re-engineered statement ingestion to support much larger files by moving parsing from the synchronous request path to Redis-backed asynchronous workers.
- Implemented distributed tracing across microservices using OpenTelemetry, propagating trace context across inter-service calls to find per-service latency bottlenecks.
- Recognised with the Uninhibited Spirit award and an innovation award for engineering ownership, cross-team collaboration, and contributions to product initiatives.

Skills Summary

- **Languages:** Python, JavaScript, Go, C++, SQL
- **Backend:** FastAPI, Django, Node.js, Express.js, REST APIs, Async I/O
- **Distributed Systems:** Microservices, Kafka, Redis, Event-Driven Architecture, Distributed Tracing, Concurrency
- **Databases:** PostgreSQL, MySQL, Oracle, MongoDB, asyncpg, ORM Design, Query Optimization, Transactions, Migrations
- **Cloud & Infrastructure:** AWS, Kubernetes, Docker, Nginx, Linux, CI/CD
- **Observability & AI:** OpenTelemetry, Prometheus, Grafana, LLM Integration, OCR, Structured Output Extraction

Education

PDPM IIITDM Jabalpur Madhya Pradesh, India

Bachelor of Technology – Information Technology 2021 – 2025

Projects

ArxivLens – Distributed Paper Ingestion & Semantic Search 2026

- Built an event-driven paper ingestion pipeline using Go, Kafka, and Python workers to crawl, parse, chunk, and embed arXiv papers into PostgreSQL with pgvector.
- Implemented idempotent processing, retry handling, dead-letter routing, and partition-keyed ordering for reliable asynchronous ingestion.
- Built a Go-based RAG query service providing citation-aware answers over indexed papers, with OpenTelemetry tracing propagated across HTTP and Kafka boundaries.

Oceanic Connectors – Async Multi-Database ORM 2026

- Built asynchronous database connectors and ORM abstractions for PostgreSQL and MySQL, providing a consistent model, query, relation, transaction, and migration interface.
- Developed a production-grade async PostgreSQL ORM on asyncpg with typed model instances, a LoopBack-style filter DSL, nested relations, JSONB fields, soft deletes, transactions, and additive migrations.
- Designed batch-based relation loading to resolve deeply nested relationships in O(depth) queries, eliminating N+1 access patterns across hasOne, hasMany, belongsTo, and hasManyThrough.

Fusion – Institute ERP (Gymkhana Module Owner) Dec 2023 – Aug 2024

- Owned end-to-end development of the Gymkhana module for a Django-based institute ERP: requirements, system design, data modelling, REST APIs, business logic, and deployment.
- Designed workflows for club management, memberships, activities, events, registrations, and role-based access, integrated with existing ERP services.
- Profiled and optimised Django ORM queries for high-traffic endpoints, reducing redundant database operations and improving API response times.

Achievements

- **Graph Theory Camp** – Placed in the top 1.25% among 40,000+ participants and earned Hall of Fame recognition.
- **Competitive Programming** – Solved 900+ algorithmic problems across competitive programming platforms.